

Introduction to Contract Theory and Auctions

Midterm II

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Screening by a Monopolist:

- You are the owner of the only pet store in town.
- A fraction λ of your customers have picky cats while the rest have carefree cats.
- You can offer a high quality cat bed and/or a low quality cat bed in your pet store.
- The following table gives the cost of offering the two qualities of cat beds as well as the willingness to pay for them by the two types of customers.

	Cost	Willingness to pay	
		Picky	Carefree
Low Quality	c_ℓ	$v_{c,\ell}$	$v_{n,\ell}$
High Quality	c_h	$v_{c,h}$	$v_{n,h}$

- $c_h > c_\ell > 0$, $v_{c,h} > v_{c,\ell} > 0$, $v_{n,h} > v_{n,\ell} > 0$.
- You choose which qualities to offer and their prices to maximize your expected profit.

- Produce both qualities.
 - Needs single crossing plus cost betweenness: $v_{c,h} - v_{c,\ell} > c_h - c_\ell > v_{n,h} - v_{n,\ell} > 0$
- Sell high quality to picky at $P_h = v_{c,h}$ and low quality to carefree at $P_\ell = v_{n,\ell}$.

- Your profit is

$$\lambda(v_{c,h} - c_h) + (1 - \lambda)(v_{n,\ell} - c_\ell)$$

- Maximizes the total surplus:
 - Extra benefit from high quality is bigger (smaller) than the extra cost for picky (carefree)

You cannot observe types

- What happens if you offer high quality at $p_h = v_{c,h}$ and low quality at $p_\ell = v_{n,\ell}$?
- Here are the net payoffs of the two types from purchasing the two qualities

	Net Payoffs	
	Picky	Carefree
Low Quality	$v_{c,\ell} - v_{n,\ell} > 0$	$v_{n,\ell} - v_{n,\ell} = 0$
High Quality	$v_{c,h} - v_{c,h} = 0$	$v_{n,h} - v_{c,h} < 0$

- Picky type wants to buy low quality.
- Everybody will buy low quality.
- Your profit will be $v_{n,\ell} - c_\ell$ only.
- Less than if you could observe types and perfectly discriminate.
- What is the best you can do in this case?

You cannot observe types

There are several things that you can do:

- ① Offer only low quality wine.
- ② Offer only high quality wine.
- ③ Offer both so that Picky chooses high and Carefree chooses low quality.

	Cost	Willingness to pay	
		Picky	Carefree
Low Quality	c_ℓ	$v_{C,\ell}$	$v_{n,\ell}$
High Quality	c_h	$v_{C,h}$	$v_{n,h}$

- What price should you set?
- If you set a price at $p = v_{n,\ell}$ both types buy.
 - Your profit is $v_{n,\ell} - c_\ell > 0$.
- If you set price at $p = v_{C,\ell}$ only type Picky consumers buy.
 - Your profit is $\lambda(v_{C,\ell} - c_\ell) > 0$.
- Therefore, your best pricing policy and resulting profit is

	Price	Profit
$\lambda \leq \frac{v_{n,\ell} - c_\ell}{v_{C,\ell} - c_\ell}$	$v_{n,\ell}$	$v_{n,\ell} - c_\ell > 0$
$\lambda > \frac{v_{n,\ell} - c_\ell}{v_{C,\ell} - c_\ell}$	$v_{C,\ell}$	$\lambda(v_{C,\ell} - c_\ell) > 0$

	Cost	Willingness to pay	
		Picky	Carefree
Low Quality	c_ℓ	$v_{C,\ell}$	$v_{n,\ell}$
High Quality	c_h	$v_{C,h}$	$v_{n,h}$

- What price should you set?
- If you set a price at $P = v_{n,h}$ both types buy.
 - Your profit is $v_{n,h} - c_h$.
- If you set price at $P = v_{C,h}$ only type C buys.
 - Your profit is $\lambda(v_{C,h} - c_h) > 0$.
- Therefore, your best pricing policy and resulting profit is

	Price	Profit
$\lambda \leq \max\{\frac{v_{n,h}-c_h}{v_{C,h}-c_h}, 0\}$	$v_{n,h}$ if $v_{n,h} - c_h > 0$, $P > v_{n,h}$ otherwise	$v_{n,h} - c_h$
$\lambda > \max\{\frac{v_{n,h}-c_h}{v_{C,h}-c_h}, 0\}$	$v_{C,h}$	$\lambda(v_{C,h} - c_h) > 0$

You offer both qualities

- You want Picky to buy high and Carefree to buy low quality cat beds.
- You want to choose P_h and P_ℓ to maximize your expected profits

$$\lambda(P_h - c_h) + (1 - \lambda)(P_\ell - c_\ell)$$

- Picky must prefer high to low quality

$$v_{c,h} - P_h \geq v_{c,\ell} - P_\ell \quad (IC_C)$$

- Carefree must prefer low quality

$$v_{n,\ell} - P_\ell \geq v_{n,h} - P_h \quad (IC_N)$$

- The above are called **Incentive Compatibility** (IC) constraints.
- And you want them to prefer to buy

$$v_{c,h} - P_h \geq 0 \quad (IR_C)$$

$$v_{n,\ell} - P_\ell \geq 0 \quad (IR_N)$$

- The above are called **Individual Rationality** (IR) (or participation) constraints.

Solving the problem

- ① (IR_N) holds with equality $P_\ell = v_{n,\ell}$.

Suppose instead that $v_{n,\ell} - P_\ell > 0$. Then

$$v_{c,h} - P_h \geq v_{c,\ell} - P_\ell > v_{n,\ell} - P_\ell > 0$$

- We could increase P_h and P_ℓ by the same amount without violating IC constraints and increase our profit.
 - This also shows that we can neglect (IR_C) .
- ② (IC_C) holds with equality: $v_{c,h} - P_h = v_{c,\ell} - P_\ell$.

Suppose instead that $v_{c,h} - P_h > v_{c,\ell} - P_\ell > v_{n,\ell} - P_\ell = 0$. Then

$$v_{c,h} - P_h > v_{c,\ell} - P_\ell > v_{n,\ell} - P_L = 0.$$

- We could increase P_H without violating any constraints and increase our profit.
- ③ Therefore, we have

$$P_\ell = v_{n,\ell}, P_h = P_\ell + (v_{c,h} - v_{c,\ell}) = v_{c,h} - (v_{c,\ell} - v_{n,\ell}).$$

- What is your profit?

$$\lambda(v_{c,h} - (v_{c,\ell} - v_{n,\ell}) - c_h) + (1 - \lambda)(v_{n,\ell} - c_\ell)$$

- Compare this with your profit under observable types.

$$\lambda(v_{c,h} - c_h) + (1 - \lambda)(v_{n,\ell} - c_\ell)$$

What is the best?

	Low Only		High Only	
	Price	Profit	Price	Profit
$\lambda \leq \max\{\frac{v_{n,h}-c_h}{v_{c,h}-c_h}, 0\}$	$v_{n,\ell}$	$v_{n,\ell} - c_\ell$	$v_{n,h}$ if $v_{n,h} - c_h > 0, P > v_{n,h}$ otherwise	$v_{n,h} - c_h > 0$ or 0
$\max\{\frac{v_{n,h}-c_h}{v_{c,h}-c_h}, 0\} < \lambda < \frac{v_{n,\ell}-c_\ell}{v_{c,\ell}-c_\ell}$	$v_{n,\ell}$	$v_{n,\ell} - c_\ell$	$v_{c,h}$	$\lambda(v_{c,h} - c_h)$
$\lambda \geq \frac{v_{n,\ell}-c_\ell}{v_{c,\ell}-c_\ell}$	$v_{c,\ell}$	$\lambda(v_{c,\ell} - c_\ell)$	$v_{c,h}$	$\lambda(v_{c,h} - c_h)$

	Both	
	Price	Profit
$\lambda \leq \max\{\frac{v_{n,h}-c_h}{v_{c,h}-c_h}, 0\}$	$(v_{c,h} - (v_{c,\ell} - v_{n,\ell}), v_{n,\ell})$	$\lambda(v_{c,h} - (v_{c,\ell} - v_{n,\ell}) - c_h) + (1 - \lambda)(v_{n,\ell} - c_\ell)$
$\max\{\frac{v_{n,h}-c_h}{v_{c,h}-c_h}, 0\} < \lambda < \frac{v_{n,\ell}-c_\ell}{v_{c,\ell}-c_\ell}$	$(v_{c,h} - (v_{c,\ell} - v_{n,\ell}), v_{n,\ell})$	$\lambda(v_{c,h} - (v_{c,\ell} - v_{n,\ell}) - c_h) + (1 - \lambda)(v_{n,\ell} - c_\ell)$
$\lambda \geq \frac{v_{n,\ell}-c_\ell}{v_{c,\ell}-c_\ell}$	$(v_{c,h} - (v_{c,\ell} - v_{n,\ell}), v_{n,\ell})$	$\lambda(v_{c,h} - (v_{c,\ell} - v_{n,\ell}) - c_h) + (1 - \lambda)(v_{n,\ell} - c_\ell)$